Greedy Algorithms for Haplotype Assembly

Rachel Ha
Mentored by Dr. Derek Aguiar
Assistant Professor of CSE
University of Connecticut

What are haplotypes?

Haplotype: A set of variations that co-occur on a strand of DNA in a chromosome

What are haplotypes?

Chromosome 1A:	C G T C G A A C T G A T
Chromosome 1B:	C G T C G A A C T G A T

What are haplotypes?

Chromosome 1A:	G G T C G A A C T G G T
Chromosome 1B:	C G C G G A A C T G A T

What are haplotypes?

Chromosome 1A:				G	G	T	C	G	A	A	C	T	G	G	T
Chromosome 1B:				C	G	C	G	G	A	A	C	T	G	A	T
genotype:	1	0	1	1	0	0	2	2	0	0	1	0			

What are haplotypes?

| | | |

What is haplotype assembly?

Haplotype Assembly: Using small DNA fragments to assemble respective haplotypes

Haplotype assembly

| | | |

Why use haplotype assembly?

- Mutations in sequences influence observable characteristics (e.g. disease).

- Genetic sequences are hard to determine experimentally and require algorithms.

Implemented Haplotype Assembly Algorithms

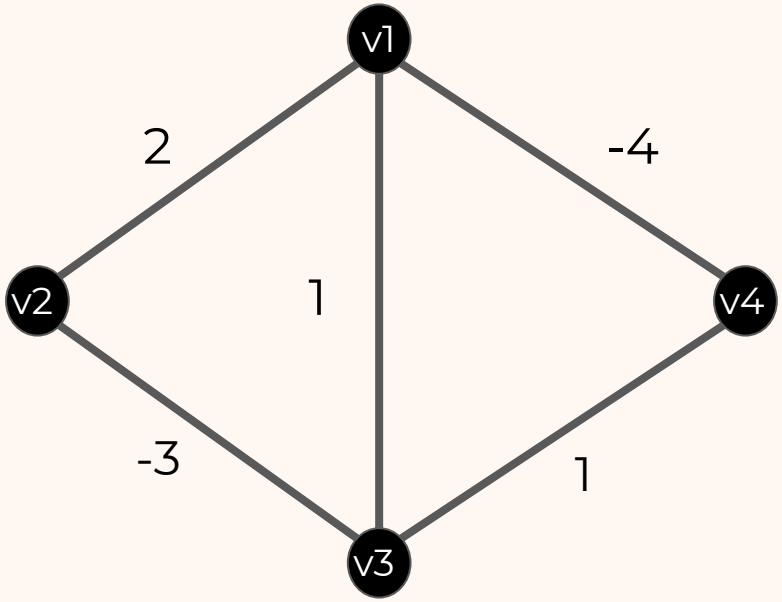
1) Hapcompass (Minimum Weighted Edge Removal)
2) Levy Algorithm

Greedy algorithm to solve MWER

- A node represents a sequence read
- An edge represents a relation between two sequence reads

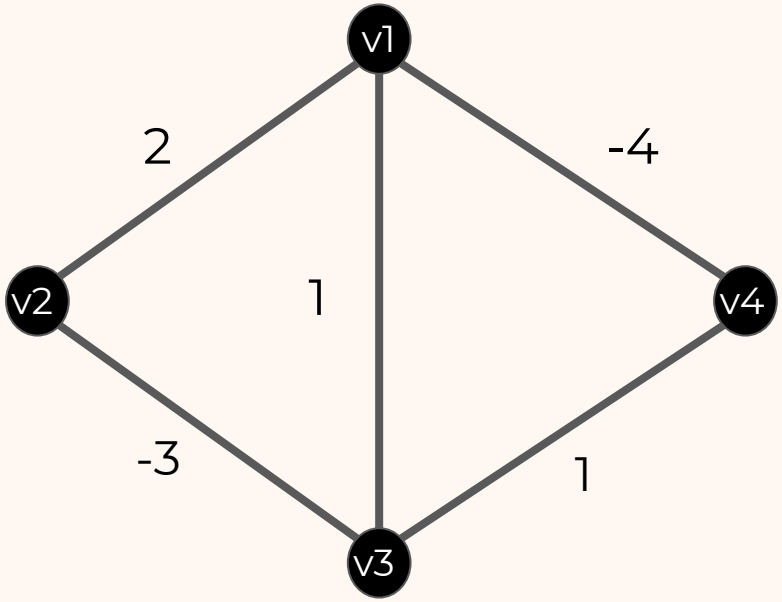
Compass Graph

- We construct a graph where
  - Vertices are variant positions
  - Edges denote the most likely phasing between two variants



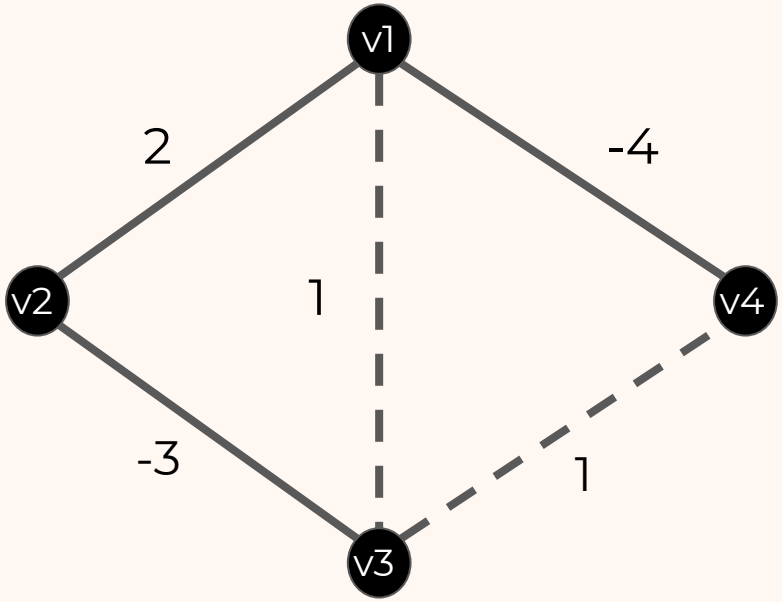
Compass Graph

- We construct a graph where
  - Vertices are variant positions
  - Edges denote the most likely phasing between two variants
- Positive numbers indicate **00** **11**
- Negative numbers indicate **01** **10**



Compass Graph

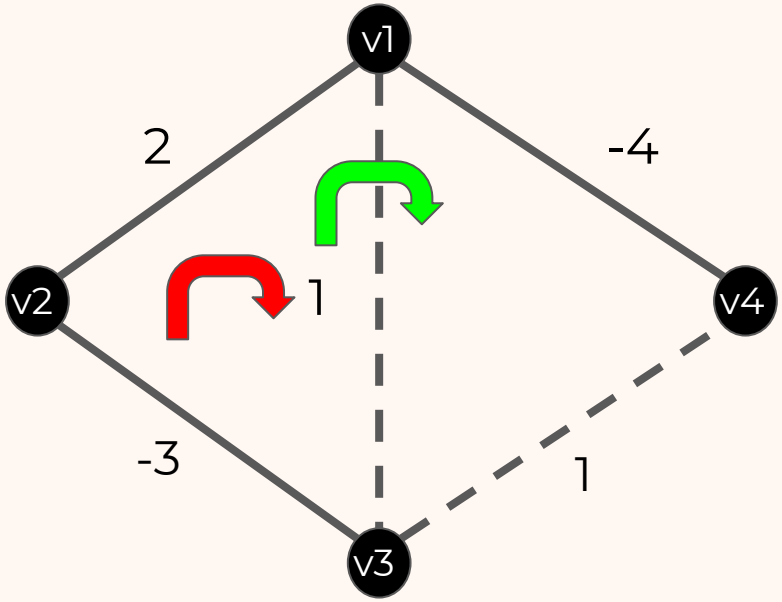
- We construct a graph where
  - Vertices are variant positions
  - Edges denote the most likely phasing between two variants
- Positive numbers indicate **00** **11**
- Negative numbers indicate **01** **10**
- First, we create a *cycle basis* by building a max spanning tree (in edge absolute value) and considering each off tree edge



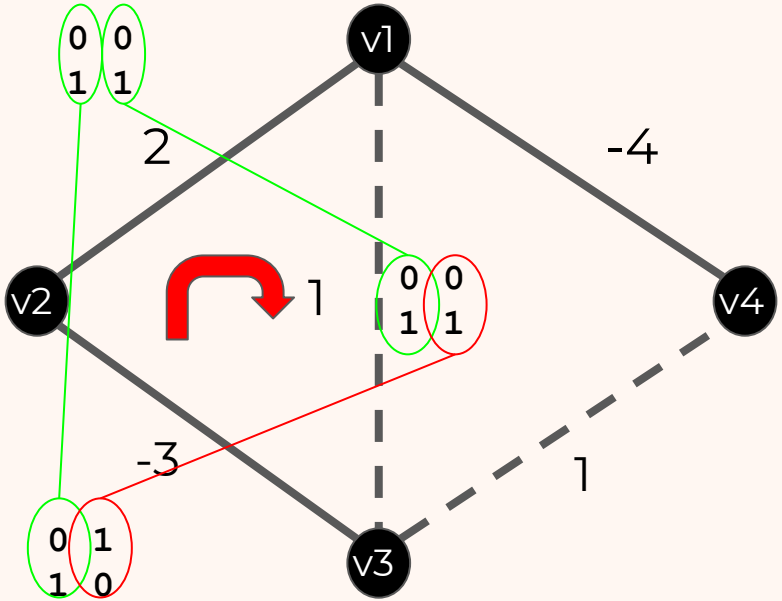
Properties of Cycles in the Compass Graph

- Inclusion of each off tree edge creates a cycle
- The two cycles here are:
  - V1-V2-V3-V1
  - V1-V2-V3-V4-V1

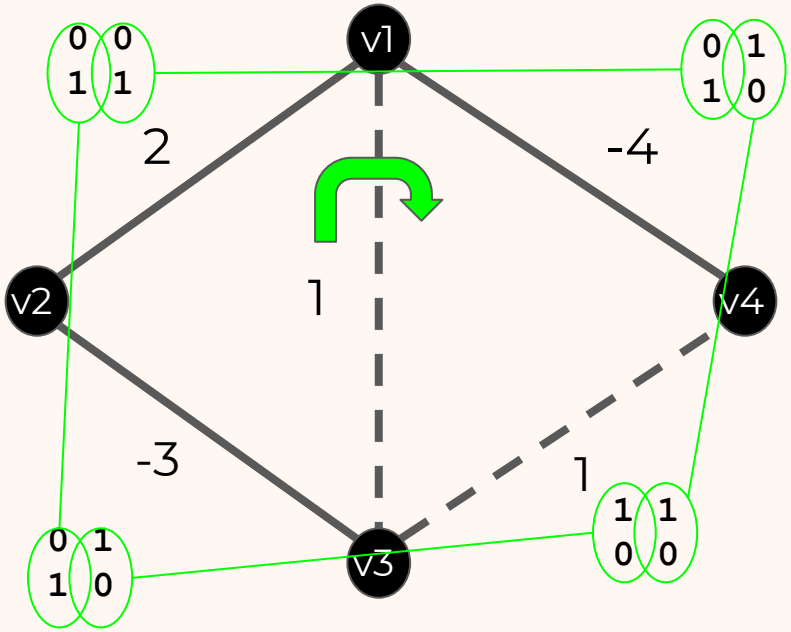
- There are two paths between any two vertices in a simple cycle and paths *define phasings*
  - If all paths in a cycle agree in the phasing, it is *happy*
  - If some paths in a cycle disagree in the phasing, it is *conflicting*



Conflicting Graph

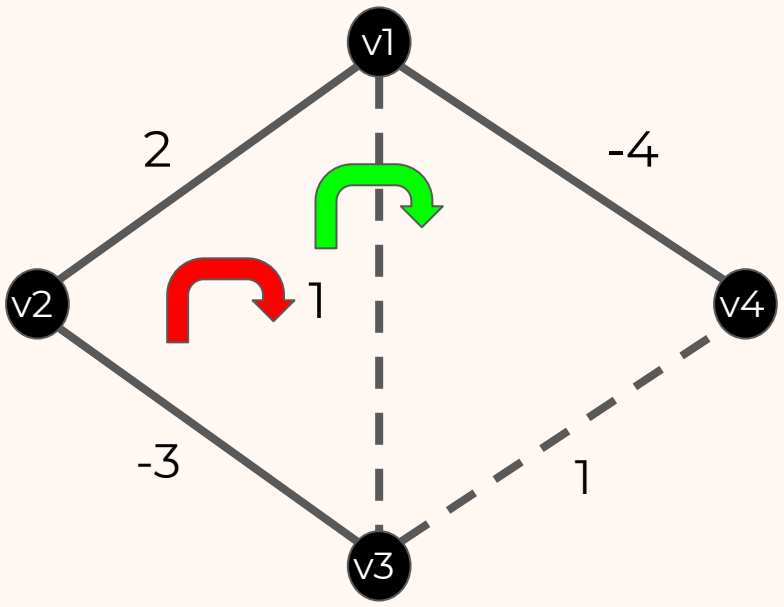


Happy Graph

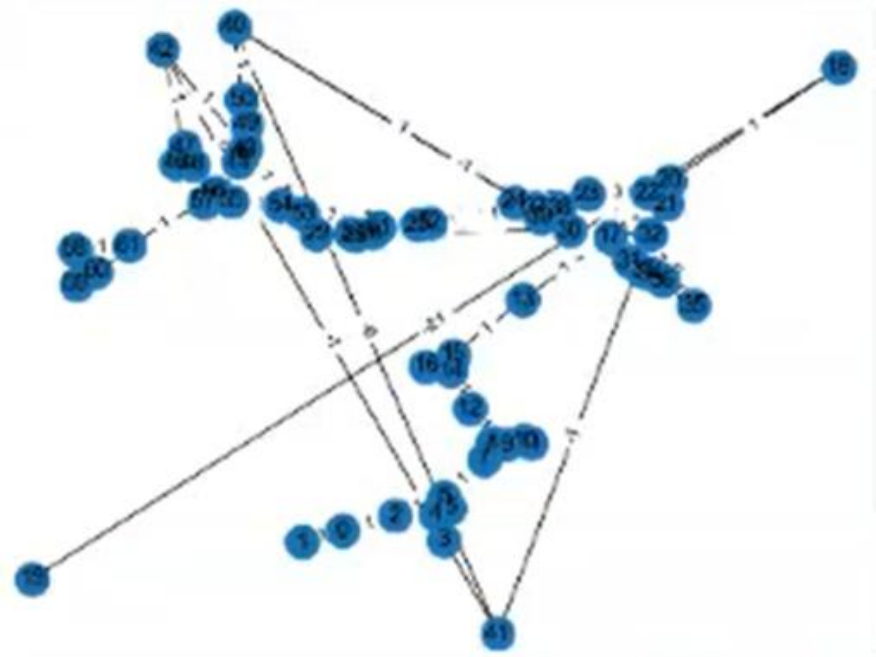


MWER Optimization Algorithm

- We implemented an algorithm in python that:
  - Finds a conflicting cycle
  - Removes the edge with the smallest weight (in absolute value) that converts the cycle into a happy cycle
  - Repeats until all cycles are happy
- A Compass graph that has no conflicting cycles has a single phasing (given by the spanning tree)



Example graph from a human sample



Levy Algorithm

Two parts:

1) Assign reads to a haplotype
2) Check each read on a majority rules basis

Levy Haplotypes

									L1			L2	L3
L4													
Chromosome 1A:				1		0	0						1
Chromosome 1B:				0		1	1						0
genotype:		1		1	1						1		

Levy Algorithm

Example Reads

	L1	L2	L3	L4
r1	1	0	0	
r2	0	1		0
r3	0	0	1	0
r4	1		0	1

Temporary Haplotypes

	L1	L2	L3	L4
Hap 1:				
Hap 2:				

Levy Algorithm

Example Reads

		L1	L2	L3	L4
#1	r1	1	0	0	
	r2	0	1		0
	r3	0	0	1	0
	r4	1		0	1

Temporary Haplotypes

	L1	L2	L3	L4
Hap 1:	1	0	0	
Hap 2:				

Levy Algorithm

Example Reads

		L1	L2	L3	L4
#1	r1	1	0	0	
#2	r2	0	1		0
	r3	0	0	1	0
	r4	1		0	1

Temporary Haplotypes

	L1	L2	L3	L4
Hap 1:	1	0	0	
Hap 2:	0	1		0

Levy Algorithm

Example Reads

		L1	L2	L3	L4
#1	r1	1	0	0	
#2	r2	0	1		0
#2	r3	0	0	1	0
	r4	1		0	1

Temporary Haplotypes

	L1	L2	L3	L4
Hap 1:	1	0	0	
Hap 2:	0	1	1	0

Levy Algorithm

Example Reads

		L1	L2	L3	L4
#1	r1	1	0	0	
#2	r2	0	1		0
#2	r3	0	0	1	0
#1	r4	1		0	1

Temporary Haplotypes

	L1	L2	L3	L4
Hap 1:	1	0	0	1
Hap 2:	0	1	1	0

Levy Algorithm

Example Reads

		L1	L2	L3	L4
#1	r1	1	0	0	
#2	r2	0	1		0
#2	r3	0	0	1	0
#1	r4	1		0	1

Temporary Haplotypes

	L1	L2	L3	L4
Hap 1:	1	0	0	1
Hap 2:	0	1	1	0

Majority Rules:

4 > 0

Levy Algorithm

Example Reads

		L1	L2	L3	L4
#1	r1	1	0	0	
#2	r2	0	1		0
#2	r3	0	0	1	0
#1	r4	1		0	1

Temporary Haplotypes

	L1	L2	L3	L4
Hap 1:	1	0	0	1
Hap 2:	0	1	1	0

Majority Rules:

2 > 1

Levy Algorithm

Example Reads

		L1	L2	L3	L4
#1	r1	1	0	0	
#2	r2	0	1		0
#2	r3	0	0	1	0
#1	r4	1		0	1

Temporary Haplotypes

	L1	L2	L3	L4
Hap 1:	1	0	0	1
Hap 2:	0	1	1	0

Majority Rules:

3 > 0

Levy Algorithm

Example Reads

		L1	L2	L3	L4
#1	r1	1	0	0	
#2	r2	0	1		0
#2	r3	0	0	1	0
#1	r4	1		0	1

Temporary Haplotypes

	L1	L2	L3	L4
Hap 1:	1	0	0	1
Hap 2:	0	1	1	0

Majority Rules:

3 > 0

Assembled Haplotypes

L1	L2	L3	L4
1	0	0	1
0	1	1	0

Assembled Haplotypes

	L1	L2	L3	L4
Hap 1:	1	0	0	1
Hap 2:	0	1	1	0

Conclusion

- Learned how to program in Python
- Implemented the HapCompass and Levy algorithms
  - Currently testing on human genome sequences
- Next Step: Implementing HapCompass for viruses

Thank you.